Agustina Díaz Peregalli

Computer Vision & Machine Learning R&D Engineer

Madrid, Spain | (Open to Relocation / Hybrid)

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PROFILE SUMMARY

Electrical Engineer and MVA Master's graduate (ENS Paris-Saclay) specializing in Computer Vision and Machine Learning. Brings 4+ years of industry R&D experience deploying high-impact CV solutions—spanning long-range multimodal vision (RGB/Thermal, VideoMAE), 3D Gaussian Splatting, and real-time edge AI. Seeking to contribute to ARQUIMEA's robotics, defense, and aerospace perception systems.

TECHNICAL SKILLS

- **3D Vision & ML/CV:** 3D Gaussian Splatting (3D GS), VideoMAE, 3DMM, PyTorch, TensorFlow, Ultralytics (YOLO), MediaPipe, OpenCV, Blender
- **Edge AI & Optimization:** Qualcomm AI Engine, TensorFlow Lite, MobileNet, Model Pruning, Quantization, Docker
- **MLOps & Cloud Platforms:** AWS (SageMaker, S3), ClearML, Google Cloud, Pandas, NumPy, Scikit-learn, Git/GitHub, Linux
- **Programming Languages:** Python, C++ (Basic), MATLAB, SQL (Basic)

WORK EXPERIENCE

Dassault Systèmes — Paris, France

Computer Vision R&D Engineer Intern — *May 2026 – Oct 2026*

- Researched and developed algorithms to estimate the internal volumetric filling of 3D Gaussian Splats (3D GS) from multi-view surface captures.
- Modeled physical material properties and structural density to enable accurate **physics-based simulation** of reconstructed 3D assets.
- *Keywords: 3D Gaussian Splatting (3D GS), Physical Parameter Estimation, Optimization, NanoBanana, VEO 3, Geometry Processing, Python.*

Digital Sense — Montevideo, Uruguay

R&D Engineer | Jr. R&D Engineer | R&D Assistant — *March 2021 – Sept 2025*

- **Multimodal Marine Vision (2025):** Architected and deployed an end-to-end multi-modal vision system (**RGB & Thermal**) for long-range marine wildlife detection. Built AWS SageMaker/S3 MLOps pipeline and trained temporal models using **VideoMAE**.
- **Edge CV & Road Inspection (2024–2025):** Developed an automated road crack and asset detection pipeline. Trained custom YOLO/Ultralytics models for adverse optical conditions and optimized latency for edge rollout via Qualcomm AI & **Docker**.
- **Real-Time Mobile CV & AR (2021–2024):** Engineered the core CV engine for a commercial virtual makeup try-on app (Ulta App) serving high-volume retail users. Fine-tuned **MobileNet** models for real-time facial landmarking and segmentation, optimized on-device inference via TensorFlow Lite pruning/quantization, and constructed fair datasets to mitigate demographic bias.
- *Keywords: Qualcomm AI, TensorFlow Lite, PyTorch, YOLO, AWS SageMaker, MobileNet, Docker, MediaPipe.*

EDUCATION

École Normale Supérieure (ENS) — Paris, France

M.S. Mathematics, Vision and Learning (MVA) — *Sept 2025 – Sept 2026*

- **Coursework:** 3D Computer Vision, Convex Optimization, Geometry Processing & Geometric Deep Learning, Probabilistic Graphical Models.
- **Honors:** Awarded Uruguay's **ANII Postgraduate Studies Scholarship** for academic excellence to fund MVA studies.

Universidad de la República — Montevideo, Uruguay

B.S. Electrical Engineering — *March 2017 – March 2024*

- **Coursework:** Machine Learning & Pattern Recognition, Image Processing, Signal Processing, Probability & Statistics.

PUBLICATIONS & FEATURED PROJECTS

- **Publication (IBERAMIA 2024 / Springer 2025):** Zimmer, S., **Díaz, A.**, Aguilera, N., Estradé, M. J., Lecumberry, F., & Musé, P. *Evaluation of Mares Uterine Health Based on Endometrial Biopsies Using Image Processing and Machine Learning Techniques.* LNCS 15277, Springer. DOI: 10.1007/978-3-031-80366-6_13
- **Kalman VAE (MVA Project):** Reimplemented *"A Disentangled Recognition and Nonlinear Dynamics Model for Unsupervised Learning"* to evaluate latent disentanglement, variational inference bounds, and nonlinear dynamics in PyTorch.

LANGUAGES

Spanish: Native | **English:** Full Professional (B2+/C1) | **French:** Conversational | **Portuguese:** Intermediate